Batch: **Roll No.: 16010124809**

Experiment No. 1

Grade: AA / AB / BB / BC / CC / CD /DD

Signature of the Staff In-charge with date

Title: Installation and Configuration of Git for Source Code Management.

Aim: To install Git and perform basic operations like clone, commit, push, and branch creation.

Tools Required: Git, GitHub, VS Code or Terminal

Theory: Git is a distributed version control system that allows teams to manage code changes over time. It helps in collaboration and source code management.

Procedure:

1. Download and install Git from https://git-scm.com/
2. Configure Git using `git config --global` commands.
3. Visit the site https://gitexercises.fracz.com/ and follow the instructions.
4. Create a local repository.
5. Add, commit, and push changes to GitHub.
6. Create and merge branches.

Expected Output: Successful commit and push of code to GitHub with branch operations.

Result: Git installed and basic source code operations performed and screenshots for each of the Exercise.

Implementation Details:

Configure git using git config –global

```
16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ git  config globaluser.name "Moksha"
```

Create local repo




```
16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~ (master)
$ cd C:/Users/KJSCE/Desktop/no.194

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194 (master)
$ cd C:/Users/KJSCE/Desktop/no.194/expt1

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ git init
Initialized empty Git repository in C:/Users/KJSCE/Desktop/no.194/expt1/.git/

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
```

Add and commit changes:

```
16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ echo "This is my first Devops expt" > README.md

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ ls
README.md

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ git add .
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.md


16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ git commit -m "Initial commit"
[master (root-commit) 329b045] Initial commit
 Committer: unknown <KJSCE@16DCEB207C-08.SVV.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 1 insertion(+)
 create mode 100644 README.md
```

Create branch b

```
16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ git branch b

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ git branch
  b
* master
```

Switch to branch b,make changes in it and commit

```
16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ git checkout b
Switched to branch 'b'

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (b)
$ echo "This is branch b line" >> README.md

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (b)
$ cat README.md
This is my first Devops expt
This is branch b line

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (b)
$ git add .
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (b)
$ git commit -m "Added chnages from branch b"
[b 30d9e1f] Added chnages from branch b
 Committer: unknown <KJSCE@16DCEB207C-08.SVV.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 1 insertion(+)
```

Merge both branches

```
16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (b)
$ git checkout master
Switched to branch 'master'

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ git merge b
Updating 329b045..30d9e1f
Fast-forward
 README.md | 1 +
 1 file changed, 1 insertion(+)

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ cat README.md
This is my first Devops expt
This is branch b line

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ git branch
  b
* master

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ echo "Back to 1st branch -1" >> README.md

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ cat README.md
This is my first Devops expt
This is branch b line
Back to 1st branch -1

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (master)
$ git switch b
M       README.md
Switched to branch 'b'

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (b)
$ cat README.md
This is my first Devops expt
This is branch b line
Back to 1st branch -1
```

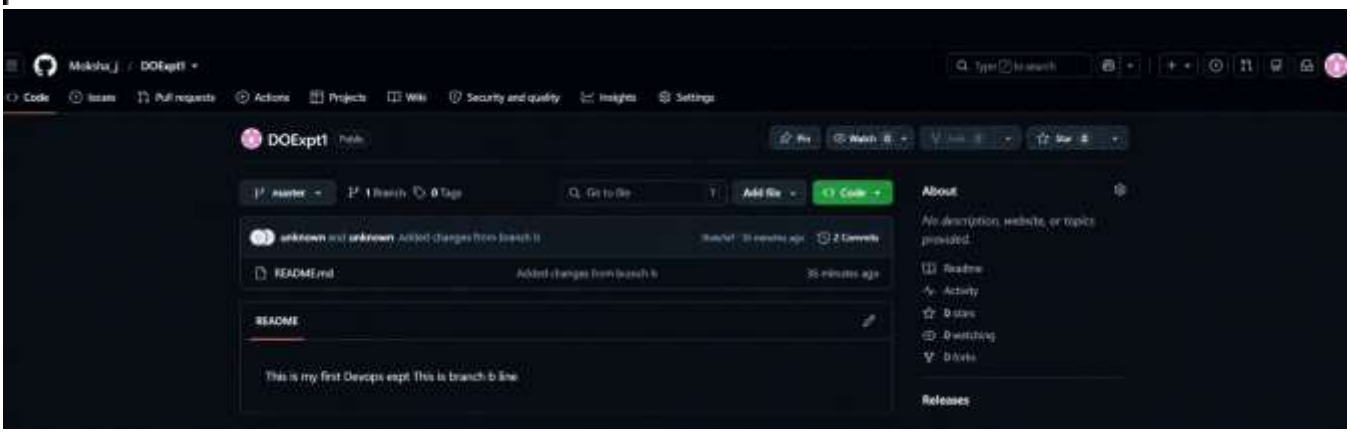
Push to github

```
16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (b)
$ git config --global user.email

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (b)
$ git remote add origin https://github.com/Mehek112/DOExpt1.git

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (b)
$ git push -u origin master
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 531 bytes | 531.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Mehek112/DOExpt1.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

16DCEB207C-08+KJSCE@16DCEB207C-08 MINGW64 ~/Desktop/no.194/expt1 (b)
$
```



Conclusion: In this experiment we have successfully use Github commands and operations.